Crop and Weed Segmentation using ResNet-Unet Architecture

Aaryan Purohit
Department of Computer Engineering
Sardar Patel Institute of Technology
Mumbai, India
aaryan.purohit@spit.ac.in

Pratik Pujari
Department of Computer Engineering
Sardar Patel Institute of Technology
Mumbai,India
pratik.pujari@spit.ac.in

Krish Shah
Department of Computer Engineering
Sardar Patel Institute of Technology
Mumbai, India
krish.shah@spit.ac.in

Dr. Anant V. Nimkar
Department of Computer Engineering
Sardar Patel Insitute of Technology
Mumbai, India
anant_nimkar@spit.ac.in

***Abstract*—The effectiveness of a ResNet-Unet (RU) architecture for semantic segmentation of soil and weeds from sorghum plants in high-resolution UAV photos is investigated. The model makes use of Unet's encoder-decoder structure for precise segmentation and ResNet's deep residual learning for efficient feature extraction. To increase the resilience of the model, data augmentation methods such random rotations, flips, and horizontal and vertical adjustments were used, along with contrast enhancement methods like Gamma Correction and Contrast Limited Adaptive Histogram Equalization (CLAHE). Utilizing the ResNet-Unet framework, we were able to attain a maximum Sørensen-Dice Coefficient (DS) of 0.929, accompanied by a standard deviation of 0.0041. Classification measures show impressive accuracy: among background, sorghum, and weed classes, precision, recall, and F1-score range from 93.01% to 99.68%. The efficiency of RU architecture for precision agriculture applications is demonstrated by this study, which also provides a viable option for targeted weed management and automated weed detection.**

***Index Terms*—Semantic Segmentation, Convolutional Neural Networks (CNN), ResNet, Unet, Deep Learning, Precision Agriculture, Weed Detection, UAV Imagery, Image Processing, Sørensen-Dice Coefficient (DS),**

I. Introduction

The global agricultural landscape faces a constant threat from weed infestations, which significantly impact crop yield and food production. Weeds, defined as unwanted wild plants that grow naturally and spread rapidly, compete with crops for essential resources such as water, sunlight, fertilizers, and soil nutrients [1]. This competition hinders growth, reduces yield, and, if uncontrolled, can lead to crop failure [2]. Traditional weed control methods, including mechanical approaches like mowing, mulching, and tilling and chemical approaches using herbicides, have proven to be labor-intensive, inefficient, and environmentally damaging [3]. The need for sustainable and efficient weed management has driven the development of site-specific weed management (SSWM). This approach involves precise application of herbicides only in weed-infested areas, minimizing herbicide usage and environmental impact [4]. Early attempts at automated weed detection relied on traditional image processing techniques and machine learning algorithms like decision trees, random forests, and support vector machines (SVM) [5].

The advent of deep learning, particularly CNNs, has revolutionized weed detection and management in precision agriculture [6]. Fully Convolutional Networks, SegNet, U-Net, and DeepLabv3 [7], have remarkably succeeded in accurately segmenting weeds from crops in various agricultural settings. Several studies have explored the application of these models for weed mapping. Huang et al. citehuang2018accurate utilized FCN with a conditional random field (CRF) for weed distribution mapping, achieving superior performance compared to traditional pixel-based methods. Among these models, U-Net has gained significant popularity for its ability to learn from relatively small datasets and its effectiveness in capturing fine-grained details, making it particularly suitable for weed segmentation tasks [8] [9]. Researchers have further enhanced U-Net's performance by incorporating modifications such as ResNet as the encoder backbone [10].

This work investigates the RU architecture's effectiveness for the precise segmentation of soil and weeds from sorghum plants in high-resolution UAV photos, addressing a critical challenge in precision agriculture. Significant contributions include the development of a robust segmentation model combining Unet's encoder-decoder structure for boundary delineation and ResNet's deep residual learning for feature extraction, demonstrating remarkable accuracy with a maximum Sørensen-Dice Coefficient (DS) of 0.929. Additionally, comprehensive exploration of preprocessing techniques, including data augmentation and contrast enhancement methods, significantly improves model robustness. Experimental results highlight the model's efficacy for practical agricultural applications, offering insights and methodologies for automated weed detection and targeted weed management, ultimately contributing to advancements in precision agriculture.

The paper is structured as follows:

In Section II, the literature survey reviews existing research relevant to the topic under investigation, providing an overview of the current state of knowledge in the field. Section III details the proposed methodology and RU architecture, accompanied by background knowledge and an illustrative architectural diagram. Following this, Section IV presents the experimental results obtained from the evaluation of the RU Model. This section scrutinizes the algorithm's performance and presents results and findings derived from the experimental analysis. Finally, Section V serves as the conclusion section, summarizing the key findings of our research.

II. Literature Survey

The past few years have seen a surge in research on leveraging deep learning for weed and crop segmentation, driven by the demand for precise, automated solutions in agriculture. CNNs have proven remarkably effective in image processing, offering significant advantages. Modern deep learning models go beyond basic segmentation by incorporating sophisticated mechanisms to enhance their performance. For example, Khan et al. [11] introduced a novel encoder-decoder framework for weed-crop segmentation in drone images. They recognized the limitations and drawbacks of the existing U-Net and FCN-based models in handling multi-scale variations and capturing contextual information. Their proposed encoder combines a Dense-inception network with the Atrous spatial pyramid pooling module to extract multi-scale features and incorporate global and local contextual details. The decoder, equipped with deconvolution layers and attention units (CnSAUs), effectively recovers spatial information and enhances the localization of crops and weeds.

Milioto et al. [12] addressed the challenge of real-time semantic segmentation in sugar beet fields by proposing a CNN-based system. Their research highlighted the difficulties in generalizing CNN classifiers to new fields due to variations in lighting, soil, and weather conditions. To overcome this, they incorporated knowledge of task-relevant background into the network to expedite training. Their system's efficacy was demonstrated on a real agricultural robot operating across different fields, showing potential for online operation in diverse environments. Haug et al. [13] presented a method for differentiating carrot from weeds using RGB and NIR images, achieving an average accuracy of 94% on a dataset of 70 images, demonstrating CNNs' effectiveness for finer-grained plant classification tasks.

Recent advancements also include incorporation of attention mechanisms into segmentation networks, enhancing accuracy by focusing on the most relevant image regions. Hu et al. [14] proposed a Channel-wise and Spatial Feature Modulation Network (CSFMN) for single image super-resolution, employing a parallel combination of channel and spatial attention units. This strategy allows for concurrent learning of spatial and channel-wise dependencies, resulting in more comprehensive feature representation. Their work demonstrated the benefits of parallel attention in capturing intricate visual patterns for enhanced image super-resolution, underscoring the potential of this approach for weed and crop segmentation tasks.

The effectiveness of different network architectures for weed and crop segmentation has also been explored. Sa et al. [15] introduced WeedMap, a large-scale semantic weed mapping framework using aerial multispectral imaging and a deep neural network. Their research focused on distinguishing vegetation from soil, showcasing the advantages of multispectral data in achieving precise segmentation. Potena et al. [16] developed a multi-step visual system for crop and weed classification using RGB+NIR imagery. They employed two distinct CNN architectures: a shallow one for vegetation detection and a deeper one for distinguishing between crops and weeds. Despite its effectiveness, the approach was limited by its pre-segmentation step, making it unsuitable for handling heavily overlapping plants, a common scenario in many agricultural settings.

To address the issue of data limitations, Wendel and Underwood [17] presented a method for generating training data for self-supervised weed detection using a multi-spectral line scanner. They mounted on a field robot, followed by vegetation segmentation and crop-row detection. Pixels within the crop rows were labeled as crops, while the remaining pixels were labeled as weeds. This self-supervised approach offered a viable solution for reducing reliance on manually labeled data, potentially simplifying the deployment of deep learning models for weed detection. Hall et al. [18] conducted a comprehensive evaluation of different features for leaf classification, comparing the performance of handcrafted and CNN features. They concluded that CNN features significantly contributed to the robustness and generalization of the classifier in challenging real-world conditions.

These studies demonstrate the growing interest and success in utilizing deep learning for weed and crop segmentation. Ongoing research continues to explore more sophisticated architectures, innovative training strategies, and efficient data utilization methods to further refine the accuracy, speed, and robustness of these solutions, paving the way for wider adoption in precision agriculture.

III. ResNet-UNet (RU Model)

In this section, we introduce the proposed methodology, ResNet-Unet (RU Model), designed to achieve precision in crop and weed segmentation. In the subsequent sections, we delve into the background information, architectural design, and process flow of RU, providing a comprehensive understanding of its implementation and functionality.

A. Background

In recent years, CNNs have exhibited remarkable performance in various computer vision tasks, including image segmentation. Among these architectures, ResNet-Unet has gained significant attention for its ability to effectively segment objects in images with high accuracy and efficiency.

1) Residual Networks (ResNet)

ResNet, introduced by He et al [19], addresses the degradation problem encountered in deep neural networks, where adding more layers deteriorates the performance. The core idea behind ResNet is the introduction of skip connections, also known as residual connections, which enable the gradient to flow directly through the network. This architecture consists of residual blocks that facilitate the learning of residual functions.

In ResNet, L represents the total number of layers, and x denotes the input to a residual block. The output of the l-th residual block, denoted as $F_l(x)$, is obtained by passing the input x through a series of convolutional layers with nonlinear activation functions, followed by the addition of the input:

$$F_l(x) = x + \mathcal{F}(x) \tag{1}$$

Here, $\mathcal{F}(x)$ represents the residual mapping learned by the convolutional layers. The skip connection ensures that the gradient can propagate through the network effectively, alleviating the vanishing gradient problem and enabling the training of deeper networks.

In ResNet, the output of the l-th residual block, denoted as H_l, is computed as follows:

$$H_l = F_l(H_{l-1}) + H_{l-1} \tag{2}$$

Where:
- H_{l-1} is the input to the l-th residual block. - $F_l(\cdot)$ represents the residual mapping learned by the convolutional layers within the l-th residual block.

2) Unet Architecture

Unet, proposed by Ronneberger et al [20]., is a popular architecture for semantic segmentation tasks. It consists of an encoder-decoder structure with skip connections between corresponding encoder and decoder layers. The encoder extracts hierarchical features from the input image, while the decoder generates a segmentation map using the features obtained by the encoder.

The encoder of Unet downsamples the input image through successive convolutional and pooling layers to capture context and spatial information. Conversely, the decoder upsamples the feature maps to generate the final segmentation map. Skip connections concatenate feature maps from the encoder with those from the decoder, allowing the decoder to access low-level details captured by the encoder.

In Unet, the encoder-decoder structure with skip connections can be represented mathematically as follows:

$$H_i = \text{Conv\_ReLU}(H_{i-1}) \tag{3}$$
$$P_i = \text{MaxPool}(H_i) \tag{4}$$

Where:
- H_i represents the feature map output of the i-th convolutional layer in the encoder. - P_i represents the feature map output of the i-th pooling layer in the encoder.

$$U_i = \text{UpConv\_ReLU}(U_{i-1}) \tag{5}$$
$$C_i = \text{Concatenate}(U_i, H_{n-i}) \tag{6}$$

Where:
- U_i represents the feature map output of the i-th upsampling convolutional layer in the decoder. - C_i represents the concatenated feature map of the i-th decoder layer with the corresponding feature map from the encoder (H_{n-i}).

The final segmentation map S is obtained by applying a convolutional layer with softmax activation at the end of the decoder.

3) ResNet-Unet Architecture

The ResNet-Unet architecture combines the strengths of both ResNet and Unet. By replacing the plain convolutional layers in the encoder of Unet with residual blocks from ResNet, ResNet-Unet enhances feature extraction capabilities and facilitates the training of deeper networks.

In ResNet-Unet, let H_l represent the output of the l-th residual block in the encoder, and G_l denote the output of the corresponding decoder block. The skip connection between the encoder and decoder layers is established by concatenating H_l with G_l, enabling the decoder to leverage rich hierarchical features extracted by the encoder.

Weed and crop segmentation require distinguishing similar plant structures using accurate location and complex feature extraction. Traditional UNet architectures excel in this due to their skip connections, preserving fine features during upsampling. However, UNet struggles to learn intricate features needed for differentiating plant species due to its shallowness.

Conversely, ResNet architectures, with their deep networks and residual blocks, excel in feature extraction but lose spatial resolution due to extensive downsampling, making them less suitable for tasks requiring precise localization. The ResNet-UNet architecture combines the strengths of both. ResNet serves as the encoder for complex feature extraction, while UNet's decoder, with skip connections, preserves spatial information. This hybrid approach ensures precise segmentation by maintaining localization and mitigating ResNet's spatial resolution loss.

B. RU architecture

In our image segmentation framework, the dataset undergoes preprocessing to ensure compatibility with the model. Initially, images of size 5472 x 3648 pixels are resized to a more manageable 256 x 256 pixels. To enhance the diversity and robustness of the training dataset, we apply several data augmentation techniques, such as horizontal and vertical flips, as well as random rotations. Additionally, techniques like brightness adjustment and noise injection are employed to simulate real-world variations in lighting and environmental conditions. The architecture of our model, illustrated in Figure 1, combines the strengths of ResNet-50 and UNet for effective segmentation, ensuring adaptability to diverse agricultural scenarios.

During the encoding phase, the preprocessed 256 x 256 input image is processed through a ResNet-50 backbone. This backbone consists of multiple residual blocks, each containing convolutional layers with a stride of two. These layers progressively reduce the spatial dimensions of the feature

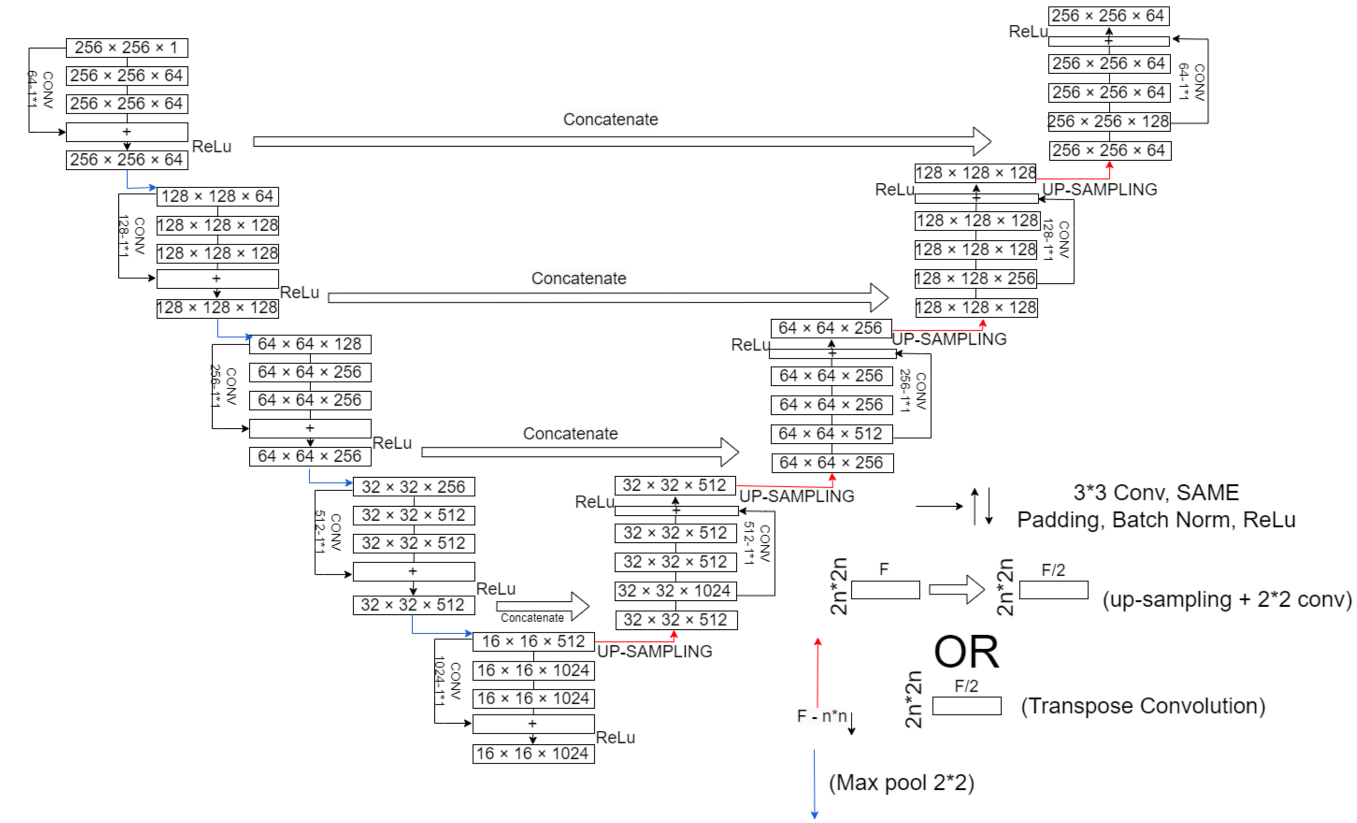

Fig. 1. Proposed RU Model Architecture. The architecture combines a ResNet-50 backbone for deep feature extraction with a U-Net structure for precise segmentation. The encoder uses ResNet-50's bottleneck blocks to capture hierarchical features, while the decoder progressively upsamples these features. Skip connections between corresponding encoder and decoder layers ensure high-resolution spatial details are preserved, facilitating accurate pixel-wise classification. The model outputs a segmentation map with the number of channels corresponding to the target classes.

maps while increasing their channel depth. For instance, the initial convolutional layer processes the three-channel (e.g., RGB) input image, generating feature maps with 64 channels and a spatial size of 128 x 128. As the image traverses through subsequent residual blocks, the feature maps are further downsampled, reaching dimensions such as 16 x 16 with 2048 channels. This hierarchical feature extraction enables the model to capture both fine and coarse details, which are critical for accurate segmentation. The residual connections within ResNet-50 ensure stable gradient flow, facilitating the training of deeper networks without degradation in performance.

The decoding phase employs the UNet architecture to upsample and segment the encoded feature maps. The decoder utilizes transpose convolutional layers, also referred to as deconvolutional layers, to incrementally increase the spatial dimensions of the feature maps. Skip connections are integrated at each stage of the decoder, concatenating feature maps from corresponding encoder layers. This mechanism ensures that both high-level semantic information and low-level spatial details are preserved. For example, the 16x16 feature maps with 2048 channels from the encoder are concatenated with the decoder's feature maps, maintaining spatial and channel consistency during upsampling. Subsequent convolutional layers refine these feature maps, ultimately producing a pixel-wise segmentation map that categorizes the input image into distinct classes. The use of batch normalization and dropout layers further enhances the model's generalization capabilities, reducing overfitting and improving performance on unseen data.

The synergy between ResNet-50 and UNet forms the backbone of our model. ResNet-50's deep residual learning framework addresses the vanishing gradient problem by incorporating shortcut connections, enabling the training of deep networks. The encoder, powered by ResNet-50, systematically downsamples the input image, capturing increasingly complex features at each stage. For instance, the feature maps evolve from 256 x 256 x 64 at the initial layer to 128 x 128 x 128 and further to 64 x 64 x 256 as they pass through successive residual blocks. This progressive reduction in spatial dimensions, coupled with an increase in channel depth, ensures comprehensive feature extraction. The encoder's ability to capture multi-scale features is complemented by the decoder's capacity to reconstruct high-resolution segmentation maps, making the model highly effective for detailed pixel-wise classification tasks.

In the decoder, the UNet architecture focuses on upsampling the feature maps. Transpose convolutions are employed to restore the spatial dimensions, while skip connections integrate feature maps from the encoder to preserve both high-resolution details and semantic context. For example, the 16 x 16 x 2048 feature maps from the encoder are concatenated with the decoder's feature maps, ensuring that fine-grained details are retained during upsampling. Convolutional layers then refine these maps, producing a final segmentation output that classifies each pixel into specific categories. The inclusion of activation functions such as ReLU and softmax ensures non-linearity and proper probability distribution across classes, respectively. This combination of techniques enables the model to handle complex segmentation tasks with high precision and recall.

To further improve image quality, advanced preprocessing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and gamma correction are applied. CLAHE enhances local contrast, making features more distinguishable, while gamma correction adjusts luminance for better visibility. These steps are followed by normalization and resizing, ensuring that the input images are standardized for efficient model training and inference. Additionally, techniques like edge detection and morphological operations are explored to highlight structural features in the images, further aiding the segmentation process. These preprocessing steps collectively ensure that the model receives high-quality input data, which is crucial for achieving optimal performance.

In summary, our model leverages the complementary strengths of ResNet-50 and UNet to achieve high-performance image segmentation. The encoder-decoder structure, enhanced by skip connections and upsampling layers, enables the model to capture both local details and global context. This integrated approach, as depicted in Figure 1, ensures accurate and reliable segmentation across diverse applications, delivering detailed and contextually aware results. The model's ability to generalize across varying conditions, combined with its robust preprocessing pipeline, makes it a powerful tool for tasks such as weed detection, crop monitoring, and precision agriculture. Future work could explore the integration of additional modalities, such as multispectral or thermal imaging, to further enhance the model's capabilities.

IV. Experimental Results

In this section, we provide an in-depth analysis of the experimental results stemming from our proposed methodology applied to the selected dataset. Our investigation encompasses various performance metrics, including precision, recall, and F1-score, alongside qualitative assessments through visualizations of ground truth and predicted masks. Through rigorous testing on distinct subsets, such as training, testing, and validation sets, we gain valuable insights into the segmentation model's efficacy in distinguishing weeds from sorghum plants in UAV imagery. Additionally, we discuss the model's adaptability to diverse growth stages and environmental conditions, offering a comprehensive understanding of its performance across different scenarios.

A. Dataset

The dataset[1] utilized in this study was derived from high-resolution aerial imagery captured over an experimental sorghum field in Southern Germany. The sorghum variety "Farmsughro 180" was cultivated with a row spacing of 37.5 cm and a seeding density of 25 seeds per m². The field also contained several dicotyledonous weed species, including Goosefoot, Field pennycress, Wild chamomile, Common gypsyweed, and Cotton thistle.

A DJI Mavic 2 Pro drone equipped with a 20 MP Hasselblad camera was employed to capture images at a resolution of 5472x3648 pixels. Flights were conducted at an altitude of 5 meters, achieving a ground sampling distance (GSD) of 1 mm, which enabled precise identification of sorghum and weeds in early growth stages (BBCH 17). A total of 60 images were collected, with 6,300 patches extracted from 19 images for annotation. Each patch was labeled pixel-wise into three categories: sorghum, weed, or soil, using the GNU Image Manipulation Program (GIMP) under the guidance of agronomy experts.

B. Results

This section evaluates the performance of a UNet architecture integrated with a ResNet-50 backbone for segmenting weeds, sorghum, and soil in UAV-captured imagery. The model was assessed using a four-fold cross-validation approach, with the optimal hyperparameters determined by averaging the Sørensen-Dice Coefficient (DS) across validation sets. The architecture achieved a DS of 0.929 with a standard deviation of 0.0041, indicating robust segmentation accuracy.

TABLE I
Classification Metrics

Class	Precision	Recall	F1-Score	Support
Background	99.80	99.93	99.86	137,909,280
Sorghum	91.58	86.10	88.76	1,249,145
Weed	87.64	72.71	79.48	574,567
macro avg	93.01	86.25	89.37	139,732,992
weighted avg	99.68	99.69	99.68	139,732,992

Figure 3 illustrates qualitative segmentation outcomes, displaying four ground patch images alongside their corresponding ground truth masks, predicted masks, and difference maps highlighting misclassifications. These visualizations demonstrate the model's ability to distinguish between weeds (orange), sorghum (blue), and background (gray). Additionally, Figure 4 presents a normalized confusion matrix, further validating the model's classification performance across the three classes.

Table I summarizes the classification metrics, including precision, recall, and F1-score for each class. The background class achieved near-perfect scores, with precision, recall, and F1-score exceeding 99%. Sorghum and weed classes also performed well, with F1-scores of 88.76% and 79.48%, respectively. The macro and weighted averages further confirm

[1]Crop and Weed Dataset: https://data.mendeley.com/datasets/4hh45vkp38/4

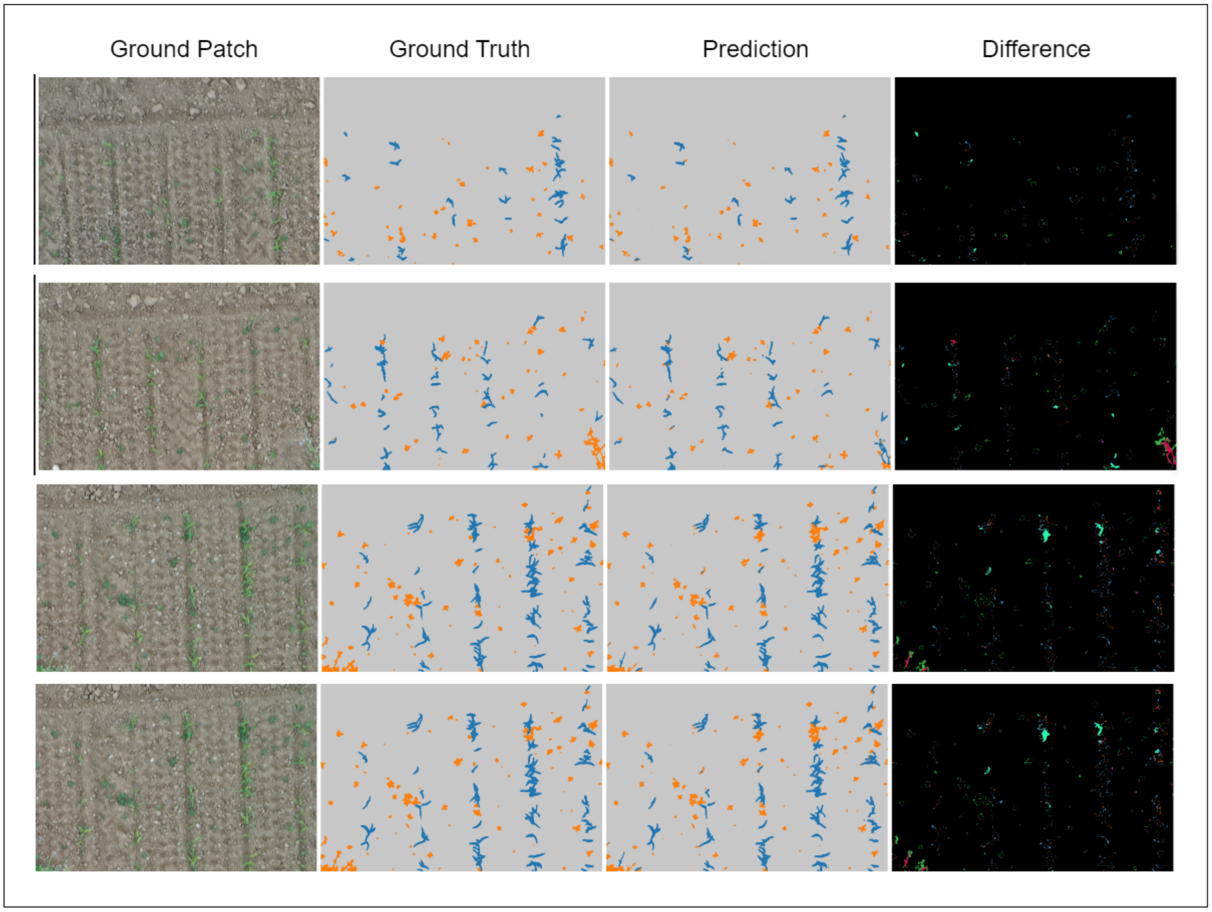

Fig. 2. Segmentation results showing 4 ground patch image and its respective ground truth mask, prediction mask and difference of ground truth mask and prediction mask generated set_1

the model's effectiveness, with metrics ranging from 86.25% to 99.68%.

TABLE II
SUMMARY STATISTICS OF THE ADDITIONAL TEST SET USED IN THIS STUDY

Test set	Patches	Sorghum	Weed
set_1	110	115	429
set_2	110	99	981

To assess generalization, two additional test sets were employed, as detailed in Table II. These sets represent distinct sorghum growth stages: set_1 corresponds to an early growth phase, while set_2 reflects a more advanced stage. The model's performance on these sets is summarized in Table III, with macro-averaged accuracy, precision, and recall consistently above 84%. Notably, the model demonstrated strong performance in detecting larger weeds, even when their size exceeded those in the training data.

However, the model exhibited a tendency to misclassify sorghum as weeds in set_1, likely due to morphological changes during growth. While partial misclassifications were common, entire sorghum plants were occasionally predicted as weeds. Additionally, edge artifacts arising from patch-based processing led to minor inaccuracies, as depicted in Figure 2. Despite these challenges, the model maintained high precision in weed detection across varying lighting conditions, underscoring its potential for real-world applications.

TABLE III
MACRO-AVERAGED RESULTS GENERATED FROM USED TEST SETS USING RESNET-UNET MODEL

Test set	Accuracy	Precision	Recall	Support
set_1 (Early stage crop growth)	0.990363	0.884283	0.820867	0.850646
set_2 (Advanced stage crop growth)	0.978746	0.854367	0.841024	0.844189

In summary, the ResNet-50 UNet architecture demonstrated strong segmentation and classification capabilities, with minor limitations in handling growth-stage variability and patch-edge artifacts. Its consistent performance across diverse datasets highlights its suitability for precision agriculture tasks.

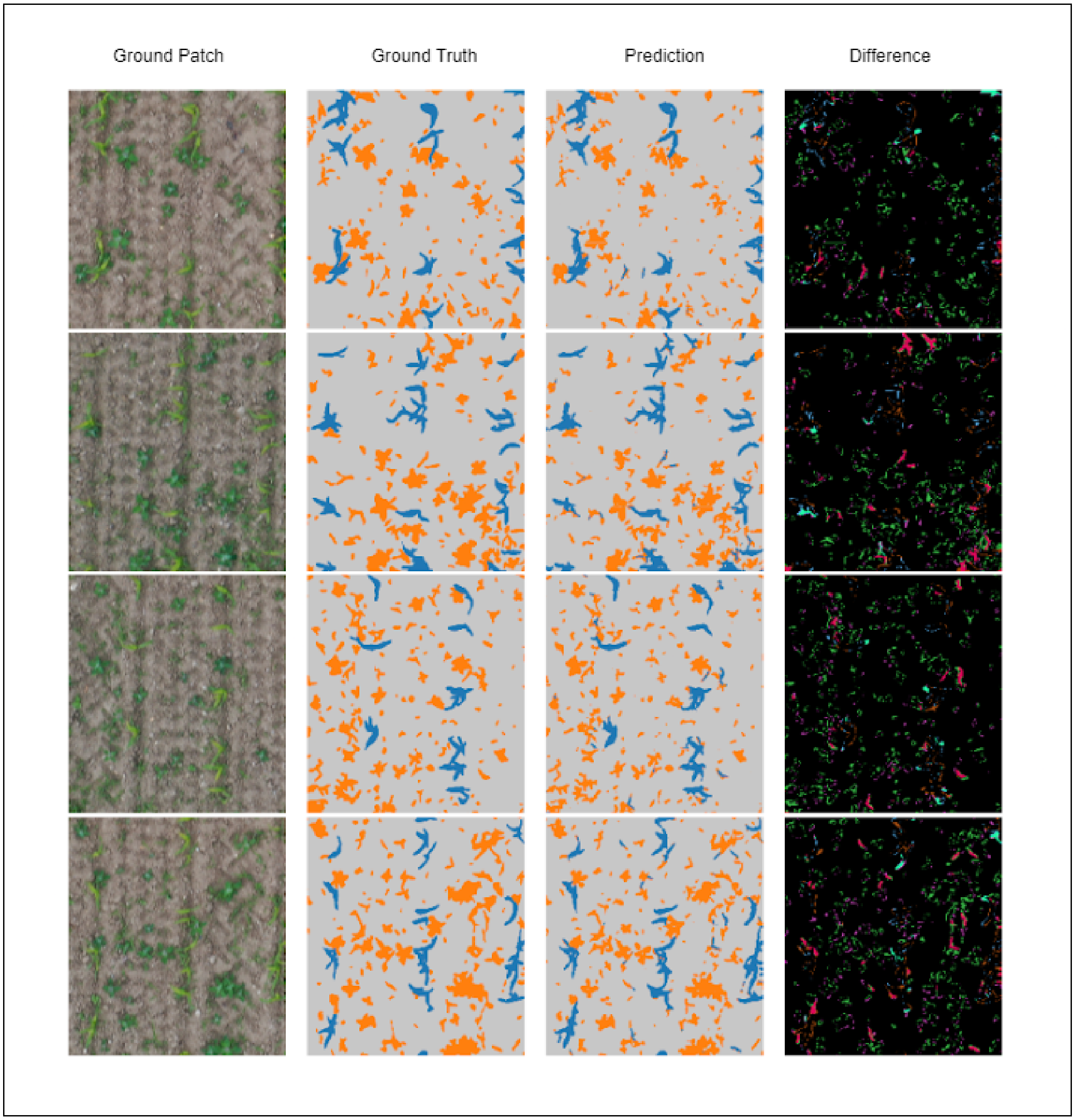

Fig. 3. Segmentation results showing 4 ground patch image and its respective ground truth mask, prediction mask and difference of ground truth mask and prediction mask generated set_2

C. Discussion

Sodjinou et al. [21] tackled the challenge of segmenting crops and weeds in complex agronomic images using a combination of semantic segmentation and K-means clustering. They employed a U-Net architecture for semantic segmentation and a subtractive clustering algorithm for K-means, achieving a high accuracy of 99.19%. The study focused on single-plant images, achieving clean and precise segmentation, particularly in scenarios where crops and weeds were visually similar.

Similarly, You et al. [22] developed a DNN-based semantic segmentation model for detecting weeds and crops, aiming to support autonomous robots in weed removal. They enhanced a Res50 backbone network with hybrid dilated convolutions, DropBlock regularization, and a novel universal function approximation block for generating color-based indices. Notably, they incorporated a bridge attention block to capture long-range contextual information and a spatial pyramid refinement block for multi-scale feature fusion. Their approach achieved state-of-the-art performance on both the Bonn and Stuttgart datasets, exceeding other prominent DNN architectures.

In contrast to these studies, our research focused on segmenting weeds from sorghum plants and soil in UAV images. We employed a Unet with a ResNet-50 feature extractor, evaluating its performance using a four-fold cross-validation

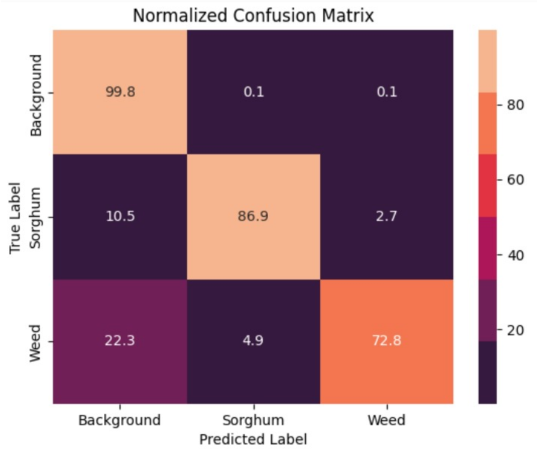

Fig. 4. Normalized Confusion matrix representing the respective classes Background, Sorghum and Weed

strategy. Our model achieved a Sørensen-Dice coefficient of 0.929, with high precision, recall, and F1-score values across all classes. Interestingly, our model demonstrated robustness to variations in illumination and could accurately predict weeds even in more advanced stages of sorghum growth where the weed size exceeded that of the training data.

However, our model did exhibit a tendency to misclassify sorghum plants as weeds in early growth stages, potentially due to morphological similarities. This suggests an area for future improvement, possibly by incorporating growth stage-specific features or refining the model's ability to distinguish subtle morphological differences. Despite this limitation, our approach offers a valuable contribution to the field of weed detection, particularly in the context of sorghum cultivation using UAV imagery.

V. Conclusion

This work highlights the great potential of the ResNet-Unet architecture for automated UAV imagery-based weed detection in sorghum fields. The model effectively separates weeds from crops, as evidenced by its high Sørensen-Dice Coefficient (DS) of 0.929 and standard deviation of 0.0041. Furthermore, classification measures support the effectiveness of the model even more; among background, sorghum, and weed classifications, precision, recall, and F1-score range from 93.01% to 99.68%. Future research directions also include developing real-time weed control systems that take advantage of the segmentation capabilities of the proposed architecture, investigating the effects of varying UAV picture resolutions, and expanding the model's applicability to additional crop types.

References

[1] A. M. Hasan, F. Sohel, D. Diepeveen, H. Laga, and M. G. Jones. A survey of deep learning techniques for weed detection from images. *Computers and Electronics in Agriculture*, 184:106067, 2021.

[2] A. Wang, W. Zhang, and X. Wei. A review on weed detection using ground-based machine vision and image processing techniques. *Computers and Electronics in Agriculture*, 158:226–240, 2019.

[3] I. Rakhmatulin, A. Kamilaris, and C. Andreasen. Deep neural networks to detect weeds from crops in agricultural environments in real-time: a review. *Remote Sensing*, 13(21):4486, 2021.

[4] H. G. Jensen, L. B. Jacobsen, S. M. Pedersen, and E. Tavella. Socioeconomic impact of widespread adoption of precision farming and controlled traffic systems in denmark. *Precision Agriculture*, 13(6):661–677, 2012.

[5] A. J. Ishak, M. M. Mustafa, N. M. Tahir, and A. Hussain. Weed detection system using support vector machine. In *2008 International Symposium on Information Theory and Its Applications*, pages 1–4. IEEE, December 2008.

[6] Z. Wu, Y. Chen, B. Zhao, X. Kang, and Y. Ding. Review of weed detection methods based on computer vision. *Sensors*, 21(11):3647, 2021.

[7] L. C. Chen, G. Papandreou, F. Schroff, and H. Adam. Rethinking atrous convolution for semantic image segmentation. *arXiv preprint arXiv:1706.05587*, 2017.

[8] K. Zou, X. Chen, Y. Wang, C. Zhang, and F. Zhang. A modified u-net with a specific data argumentation method for semantic segmentation of weed images in the field. *Computers and Electronics in Agriculture*, 187:106242, 2021.

[9] L. Hashemi-Beni and A. Gebrehiwot. Deep learning for remote sensing image classification for agriculture applications. *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLIV-M-2–2020:51–54, 2020.

[10] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[11] SD Khan, S Basalamah, and A Lbath. Weed–crop segmentation in drone images with a novel encoder–decoder framework enhanced via attention modules. *Remote Sensing*, 15(23):5615, 2023.

[12] A Milioto, P Lottes, and C Stachniss. Real-time semantic segmentation of crop and weed for precision agriculture robots leveraging background knowledge in cnns. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2229–2235. IEEE, 2018.

[13] H. Huang, Y. Lan, J. Deng, A. Yang, X. Deng, L. Zhang, et al. Accurate weed mapping and prescription map generation based on fully convolutional networks using uav imagery. *Sensors*, 18(10):3299, 2018.

[14] Y Hu, J Li, Y Huang, and X Gao. Channel-wise and spatial feature modulation network for single image super-resolution. *IEEE Transactions on Circuits and Systems for Video Technology*, 30(11):3911–3927, 2020.

[15] I Sa, M Popović, R Khanna, Z Chen, P Lottes, F Liebisch, and R Siegwart. Weedmap: a large-scale semantic weed mapping framework using aerial multispectral imaging and deep neural network for precision farming. *Remote Sensing*, 10(9):1423, 2018.

[16] C Potena, D Nardi, and A Pretto. Fast and accurate crop and weed identification with summarized train sets for precision agriculture. In *International Conference on Intelligent Autonomous Systems*, pages 105–121. Springer, 2016.

[17] A Wendel and JP Underwood. Self-supervised weed detection in vegetable crops using ground based hyperspectral imaging. In *2016 IEEE International Conference on Robotics and Automation (ICRA)*, pages 3556–3563. IEEE, 2016.

[18] D Hall, CS McCool, F Dayoub, N Sunderhauf, and B Upcroft. Evaluation of features for leaf classification in challenging conditions. In *Proceedings of the IEEE Winter Conference on Applications of Computer Vision*, pages 797–804, 2015.

[19] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. pages 770–778, 06 2016.

[20] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. volume 9351, pages 234–241, 10 2015.

[21] S. G. Sodjinou, V. Mohammadi, A. T. S. Mahama, and P. Gouton. A deep semantic segmentation-based algorithm to segment crops and weeds in agronomic color images. *Information Processing in Agriculture*, 9:355–364, 2022.

[22] J. You, W. Liu, and J. Lee. A dnn-based semantic segmentation for detecting weed and crop. *Computers and Electronics in Agriculture*, 178:105750, 2020.